Macroeconomics – Measuring A Nation's Income (TOPIC 1/ Chapter 23)

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Objectives and Instruments of Macroeconomics

- Objectives:

Output: High level and rapid growth of output

Employment: High level of employment with involuntary unemployment

Price-level stability

Instruments of Macroeconomics

- Instruments

  Monetary policy:

  Controlling the money supply to determine the interest rates

  Fiscal Policy:

  Government expenditure

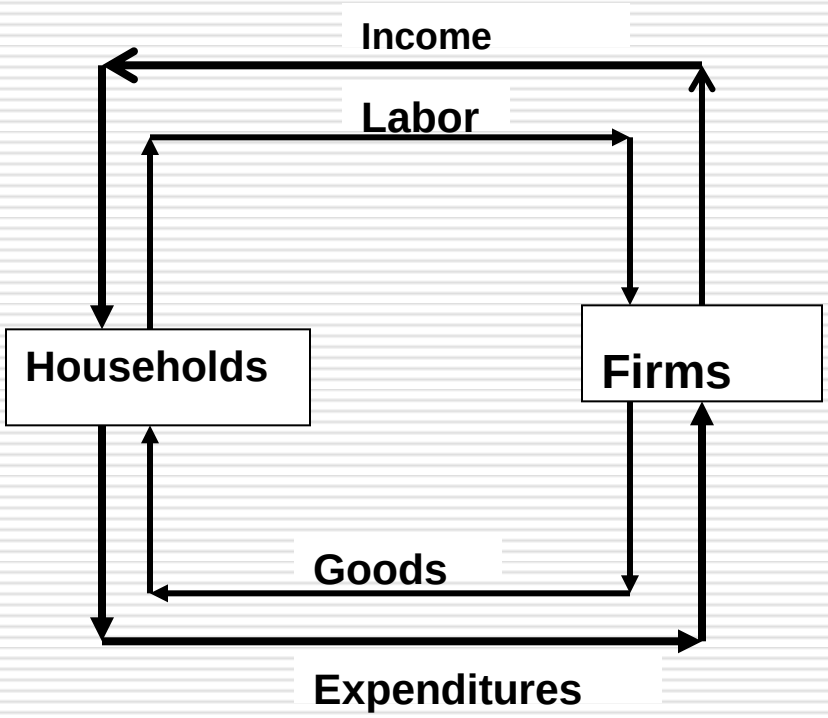
  Taxation

Measuring National Income

- *Three main statistics of macroeconomics*:

1. **GDP**—Gross domestic products, which tells us the nation's total income and the total expenditure on its output of goods and services
2. **CPI**—the Consumer Price Index, which measures the level of prices (inflation)
3. **Unemployment rate** —this tells the fraction of workers who are unemployed.

Circular Flow



This figure illustrates the flows between firms and households in an economy that produces one good bread from one input labor. The inner loop represents the flows of labor and bread; household sell their labor to firms, and the firm sell the bread they produce to household.

The outer loop represents the corresponding flows of dollars; household pay the firms for the bread, and the firms pay wages and profit to the households. In this economy GDP is both the total expenditure on bread and the total income from the production of bread.

Gross Domestic Product (GDP)

Gross Domestic Product (GDP)

It is the market value of all final goods and services produced within a country in a given time period

Gross Domestic Product (GDP)

- It measures the value of economic activity
- This gives an idea how well an economy is performing
- **Two way of explanation**:

--*the total income of everyone in the economy*

--*the total expenditure on the economy's output of goods and services*

GDP (contd.)

- Three methods of GDP calculation:

1. Income method —GDP is the total income from the production of goods and services, which equals the sum of wages and profit

GDP = Compensation of employees **+** rents **+** profits **+** net interest

2. Expenditure method —GDP is also the total expenditure on purchases of goods and services

GDP calculation

- Value added method —GDP is also the total value added of all firms in the economy. The *value added* of a firm equals the value of the firm's output *minus* the value of the intermediate goods that the firm purchases

Value added method

- The market value of a firm's product or service minus the cost of inputs purchased from other firms
- The summing of the value added by all firms in the economy yields the total value of final goods and services

Value added method (Table)

Firm	Revenue	− Cost of intermediate goods	= Value added
Grain supplier	$0.50	$0.00	$0.50
Flour miller	$1.20	$0.50	$0.70
Bakery	$2.00	$1.20	$0.80
Total			$2.00

- Each stage of production adds value
- But if we add up the total sales made at all stages of production we will double count intermediate inputs
- Value of final sales = the total value added at all stages of production
- GDP sums up value added at each stage of production.

GDP calculation—Some rules

- *GDP is the market value of all **final goods and services** produced within an economy in a given period of time.*
- *An economy produces many goods and services. How the above definition applies, we'll see next.*

GDP Calculation-Rules

1. To compute the total value of different goods and services, the national income accounts use market prices because the prices reflect how much people are willing to pay for a good or service.

GDP = (Price of apple * Quantity of apple)+ (Price of orange* quantity of orange)+…….

GDP calculation—Rules

2. *Used goods*: When a company produce something, its sells value is added. But when a company transfers a produced item to another company, it will not be added.

The sale of used goods is not included in the GDP

GDP calculation—Rules

- *Inventory*: When a firm increases its inventory of goods, this investment in inventory is counted as expenditure by the firm owners. Thus production for inventory increases GDP.
- A sell out of inventory does not influence the GDP

GDP calculation—Rules

- *Intermediate goods and value added*:

  GDP includes only the value of final goods

  To add the intermediate goods will lead to double counting

GDP calculation—Rules

- *Imputations*: Some goods and services are not sold in the market. For these we need to estimate the value which is called imputed value.

- Imputations are specially important for determining the value of housing.

- For homeowners, rent is calculated considering market rent
- This is also necessary for govt. services.
- No imputation is necessary for *underground economy*.

GDP & Components of Expenditure

- The national income accounts divide GDP into four broad categories of spending:

1. Consumption(C)
2. Investment(I)
3. Government purcheases(G)
4. Net export(NX)

Thus, GDP= C+I+G+NX

GDP & Components of Expenditure

GDP

Consumption

- Nondurable goods
- Durable goods
- Services

Investment

- Nonresidential fixed investment
- Residential fixed investment

Government Purchases

- Defense
- Non-defense

Net Export

- Export
- Import

Other Measures of Income

- The national income accounts include other measures of income that differ slightly in definition from GDP
- GNP= GDP + Factor Payments From Abroad- Factor Payments to Abroad
- GNP measures the total income earned by nationals whereas GDP measures total income produced domestically.

Other Measures of Income

- NNP= GNP – Depreciation
- National Income = NNP – Indirect Taxes

  National Income measures how much everyone in the economy has earned.

Personal Income:

- A series of adjustments takes us from national income to personal income, the amount of income that households and non-corporate businesses receive.

- Personal Income= National Income - corporate profits – Social insurance contributions – Net Interest + Dividends + Governments Transfers to Individuals + Personal Interest Income.

Personal Income:

- Disposable Personal Income =

  Personal Income – Personal Tax & Non-tax Payments

  We are interested in disposable income because it is the amount households and non corporate businesses have available to spend after satisfying their tax obligations to the government

Real GDP vs Nominal GDP

- The value of goods and services measured at current prices is the **nominal GDP.**
- The value of goods and services measured at constant prices (a *base-year*) is the **real GDP.**
- Real GDP is the better measure of economic well-being

Real GDP

- In calculating real GDP, we use a base-year price of commodities.
- Because the prices are held constant, real GDP varies from year to year only if the quantities produced vary.
- Because a society's ability to provide economic satisfaction for its members ultimately depends on the quantities of goods and services produced, real GDP provides a better measure of economic well-being.

Real GDP

- **GDP Deflator**

  It is defined as the ratio of nominal GDP to real GDP:

  GDP deflator: (**Nominal GDP/Real GDP)100**

- The GDP deflator reflects what's happening to the overall level of prices in the economy

GDP deflator

- **If P is the price of bread and Q is the quantity sold, then nominal GDP is P*Q.**
- **Real GDP = $P_{(base)*}$ Q**
- **GDP deflator: P/P_{base}**
- The definition of GDP deflator allows us to separate nominal GDP into two parts: one part measures quantity (real GDP) and another part measures prices (GDP deflator)
- Nominal GDP = Real GDP * GDP deflator

GDP AND ECONOMIC WELL-BEING

- GDP is the best single measure of the economic well-being of a society.
- GDP per person tells us the income and expenditure of the average person in the economy.

GDP AND ECONOMIC WELL-BEING

- Higher GDP per person indicates a higher standard of living.
- GDP is not a perfect measure of the happiness or quality of life, however.

GDP AND ECONOMIC WELL-BEING

- Some things that contribute to well-being are not included in GDP.
  - The value of leisure.
  - The value of a clean environment.
  - The value of almost all activity that takes place outside of markets, such as the value of the time parents spend with their children and the value of volunteer work.

Table 3 GDP, Life Expectancy, and Literacy

Country	Real GDP per Person (1999)	Life Expectancy	Adult Literacy
United States	$31,872	77 years	99%
Japan	24,898	81	99
Germany	23,742	78	99
Mexico	8,297	72	91
Russia	7,473	66	99
Brazil	7,037	67	85
China	3,617	70	83
Indonesia	2,857	66	86
India	2,248	63	56
Pakistan	1,834	60	45
Bangladesh	1,483	59	41
Nigeria	853	52	63

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